

CS 300 | Advanced Computer Graphics I

Assignment 4 | Reflection and Refraction with Cube Mapping

Description

For this programming assignment you need to implement reflection and refraction system using cube mapping. The parsed scene format will have all the content used in assignment 0 with new added data listed below, but some of the features implemented will not be used.

- Light data:
 - No light data will be provided for this assignment.
- Object data:
 - `reflector`: only one object in the scene should have this attribute and it will be the one that will reflect/refract the environment surrounding it. Along with the attribute and index of refraction will be parsed. Default value will be 1.
- Environment data:
 - `Environment`: list of 6 textures in a specific order representing the environment cube map to apply to the skybox. Order of the textures will be: right, left, top, bottom, front and back.

The application should:

1. Read the input file
 - a Load/Generate mesh data
 - b Generate Texture
2. Upload data to OpenGL
3. Accept from the keyboard the user input in order to move the camera
4. Update objects with the corresponding animation
5. Build the camera and perspective transformation according to the reflector object
6. Set frame buffers, shader program and uniform to render shadow map
7. Draw skybox.
 - a **Vertex Shader:**
 - i Transform position to NDC space.
 - ii Send model space position to fragment shader.
 - b **Fragment Shader:**
 - i Sample the skybox environment map using the model space position.
8. Draw each mesh (first pass):
 - a **Vertex Shader:**
 - i Transform position to NDC space.
 - b **Fragment Shader:**
 - i Draw the object with texture.
9. Set frame buffer, shader program and uniforms to render the scene from the camera
10. Draw skybox.
11. Draw each mesh
 - a **Vertex Shader:**
 - i Transform normal and position from model to world space.

- ii Transform position to NDC space.
 - b **Fragment Shader:**
 - i Compute view vector in world space and the reflected/refracted vector.
 - ii Sample the cube map with the computed vector.
12. Finish frame

Notes

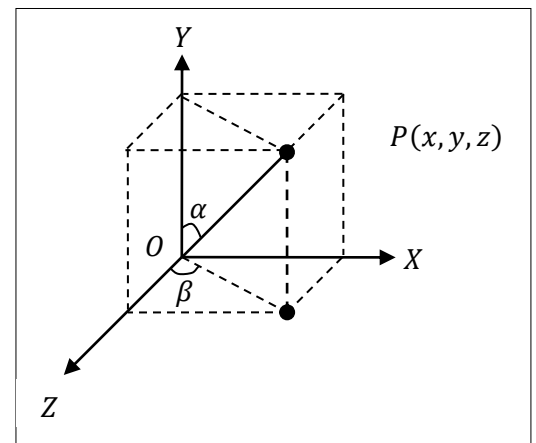
- Lighting mapping is **NOT** required for this assignment and it should be disabled.
- Normal mapping is **NOT** required for this assignment and it should be disabled.
- Shadow mapping is **NOT** required for this assignment and it should be disabled.

Material Properties

- The checkerboard texture is the material applied to every object except for the reflective/refractive one

Input

- Camera will be controlled using spherical coordinates defined in the diagram. The target will be the origin of that coordinate system. The input should alter r , α and β as follows:
 - W: Make α angle smaller. It should move the camera towards the top of the target.
 - S: Make α angle greater. It should move the camera towards the bottom of the target.
 - A: Make β angle smaller. It should move the camera towards the left rotating around the target.
 - D: Make β angle greater. It should move the camera towards the right rotating around the target.
 - E: Make r greater. It should move the camera away from the target.
 - Q: Make r smaller. It should move the camera closer to the target.
- N: Toggle normal/tangent/bitangent rendering
- T: Cycle between the different modes (texture/reflective/ refractive)
- F: Toggle face/averaged normal
- M: Toggle wireframe on/off
- +/- or Z/X: Increase/reduce number of slices (4 is the minimum number of slices)



Assignment Submission

Please refer to the syllabus for assignment submission guideline. Failure to the submission guidelines correctly might cause you to lose point.

This assignment will have a different late submission guideline. Submitting before **2 days** after the deadline will be a **-5% penalty** and if the submission happens before **6 days** after the deadline the penalty will be of **-20%**. No submission will be grade after those 6 days.

Grading Rubrics

The following is a rough guideline on how your assignment will be graded and the weight of each part.

Feature	Grade %
Sky box rendering	15%
Environment map generation	35%
Environment map application	40%
Scene and presentation	10%